Final Exam

Course Title	**Microprocessor systems lab**		**Course Code**	**ELE0328**	
Semester	**Second**		**Academic Year**	**2023/2024**	
Exam Date	**27/05/2024**	**Day**	**Monday**	**Time**	**8:30-9::30**

Student Name		**Student ID No.**		**Section**	

GENERAL INSTRUCTIONS

1. Students are.not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Essay	CLIO 1-7	18(72%)			
2	Essay	CLIO 1-7	7(28%)			
Total			25(100%)			

Moderation Review			
Comments			
Marks Reviewed by		**Signature & Date**	

Problem # 1: Write an 8086 assembly code for the following:

A) Print out on the console screen **main_menu. [2 PTS]**

B) Print out on the console screen **MSG1,** allow the input of a character (A, S, C, or D), and then store it in **choice**. **[4 PTS]**

C) Print out on the console screen **MSG2,** allow the input of first number between 0 and 9, and then store it in **number1**. **[4 PTS]**

D) Print out on the console screen **MSG3,** allow the input of the second number between 0 and 9, and then store it in **number2**. **[4 PTS]**

E) Perform the arithmetic operation on the two numbers stored on **number1** and **number2** and then store the result on **result**. **[4 PTS]**

```
ORG 100h

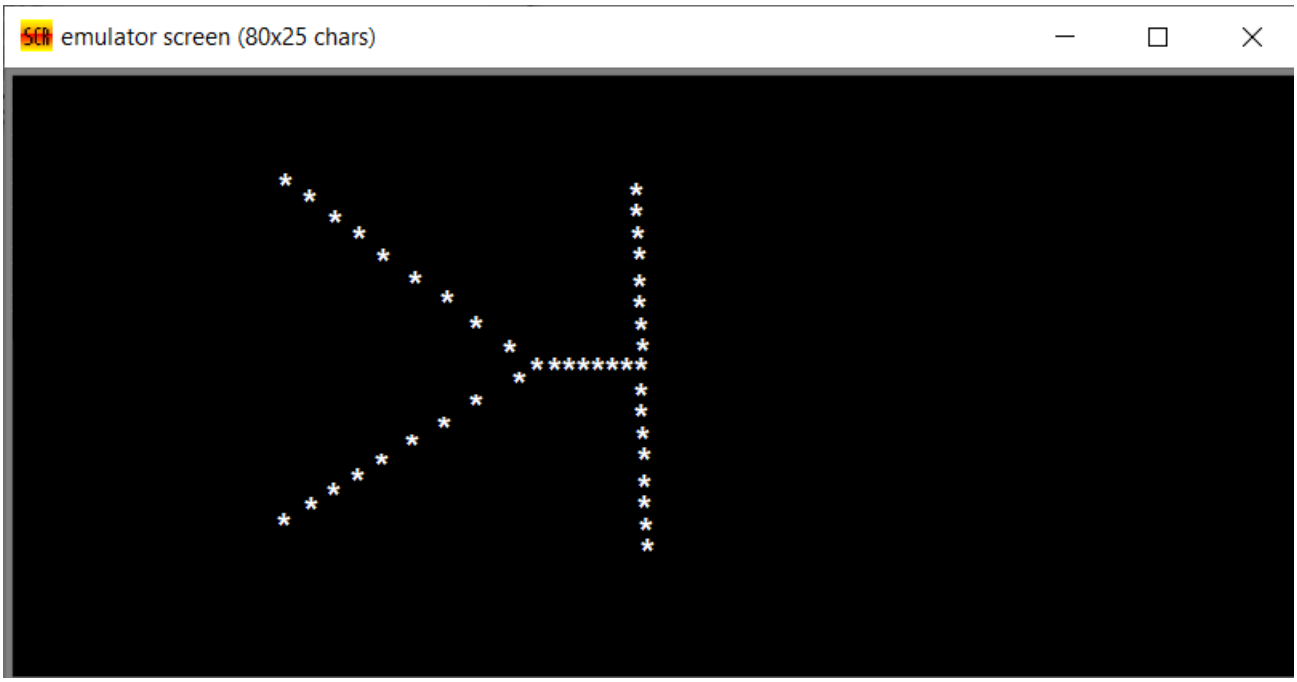
HLT

main_menu db "*****************************",0dh,0ah
          db "* Enter A for Addition ",0dh,0ah
          db "* Enter S for Subtraction ",0dh,0ah
          db "* Enter D for Division ",0dh,0ah
          db "* Enter M for  Multiplication ",0dh,0ah,
          db "* Enter B for  setting background and foreground colors to cyan and light
red respectively  ",0dh,0ah,
          db "* Enter D to Clear screen and then display a pattern in screen
middle",0dh,0ah,
          db "*****************************",0dh,0ah, ,0dh,0ah,'$'

MSG1 db   0dh,0ah,"Enter your Choice:",0dh,0ah,'$'
MSG2 db   0dh,0ah,"Enter First number:",0dh,0ah,'$'
MSG3 db   0dh,0ah,"Enter Second number:",0dh,0ah, 0dh,0ah,'$'

choice db ?
number1 db ?
number2 db ?
result db ?
```

Pattern to Draw:



Problem # 1: Write an 8086 assembly code to find the sum of all numbers in byte memory array named "**X**" that are divisible by 5. **[7 PTS]**

```
ORG 100H


HLT
X db 4, 6, 7, 8, 9, 15, 23, 30, 45, 27, 14
```

Good Luck